

# IMPERIAL

## ADAPTIVE ACQUISITION: A REINFORCEMENT LEARNING FRAMEWORK FOR AUTONOMOUS MRI SEQUENCE OPTIMIZATION

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An Interim Report submitted in fulfillment of requirements for the degree of  
**Bachelor of Engineering in Computing**

Department of Computing  
Imperial College London  
2026



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# 1

## Introduction

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### 1.1 Introduction

In 2021, there were 395,000 cases of cancer in the UK, with 1 in 2 people in the UK being diagnosed with cancer in their lifetime. [1] Radiotherapy (RT) is a crucial component of cancer treatment; indeed, it is required for more than 50% of cancer patients. [2]

Image-guided radiation therapy (IGRT) is used to increase the precision of treatment and ensure accurate dose delivery. Conventionally, images are taken using a high-resolution X-ray technique called Cone Beam Computed Tomography. This is still the most common technique. [3]

However, poor image quality (caused by excessive scattering of X-ray photons) and ionising radiation are key drawbacks. Instead, Magnetic Resonance Imaging (MRI) can be used to improve soft tissue contrast (vital for liver or pancreatic cancer) and increase the accuracy of tumor target delineation. MR-guided radiation therapy (MRgRT) has been shown to significantly improve patient prognosis. [3]

MRI and a linear accelerator (the source of radiation) have been coupled into a hybrid Magnetic Resonance Accelerator (MR-Linac), the latest innovation in IGRT. The MR-Linac allows on-table adaptation of treatment plans, to respond to tumour movements and progression. However, one

of the clinical bottlenecks is the extra time required for image acquisition. [4] The high-resolution images needed for accurate delineation require extended scan time. This also increases the risk of motion artifacts (inaccuracies due to patient movement).

There has been significant research into accelerating image acquisition, focusing mainly on using Deep Learning (computational neural network approaches) for image reconstruction (translating the raw MR signal into an image). Leading to significant improvements to image robustness, artifact reduction and image quality when using undersampled data.

There is now a growing interest in using DL and Reinforcement Learning (agent-based approaches) to develop novel pulse sequences, which are currently designed by MR experts. Novel pulse sequences have shown promise in speeding up image acquisition while maintaining image quality. [5]

# 2

## Background & Related Work

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It is necessary to understand the current process of MRI Pulse Sequence design before attempting to use DL or RL to improve upon it. This requires an understanding of MRI-Physics; a short introduction follows.

### 2.1 Principles of Magnetic Resonance Imaging

The signal generated by an MRI uses the Nuclear Magnetic Resonance (NMR) of protons. All subatomic particles have intrinsic angular momentum, which is a quantum property called spin.

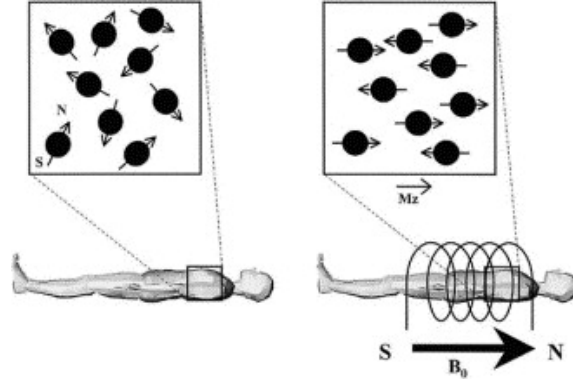


Figure 2.1: Protons align in parallel or antiparallel to the external field. Parallel alignment is a lower energy state, thus slightly more popular, leading to a net magnetisation vector ( $M_z$ ). Figure from [7]

For example, a proton has a spin- $\frac{1}{2}$ , while a neutron has a spin- $\frac{1}{2}$ . For a nucleus to have a net spin, it cannot have the same number of neutrons and protons.

A Hydrogen nucleus is made of a single proton, so it has a net spin- $\frac{1}{2}$ . The angular momentum, combined with the mass and charge of a proton, results in a magnetic moment  $\mu$ . When a uniform magnetic field ( $B_0$ ) is applied, protons will align their magnetic axis with the field. (See Figure 2.1) However, a proton wobbles around the field's axis, which is called precession. Essentially, the proton behaves like a microscopic bar magnet rotating around its axis.[6]

The frequency of this rotation is known as the Larmor frequency ( $\omega_0$ ) and is directly proportional to the magnetic field ( $B_0$ ). The constant of proportionality is called the gyromagnetic ratio ( $\gamma$ ). For protons,  $\gamma/2\pi = 42.58 \text{ MHz T}^{-1}$ . [7]

$$\omega_0 = \gamma B_0 \quad (2.1)$$

### 2.1.1 RF Pulses

The overall net magnetisation effect from all protons is a vector,  $M$ . This vector can be broken down into an x-y transverse magnetisation vector ( $M_{xy}$ ) caused by the precession effect and the magnetisation vector aligned with the  $B_0$  field ( $M_z$ ).

The transverse magnetisation induces a signal (at the Larmor Frequency) in a pair of receiver coils placed perpendicular (as seen in Figure 2.2). To measure  $M_z$ , a second perpendicular magnetic



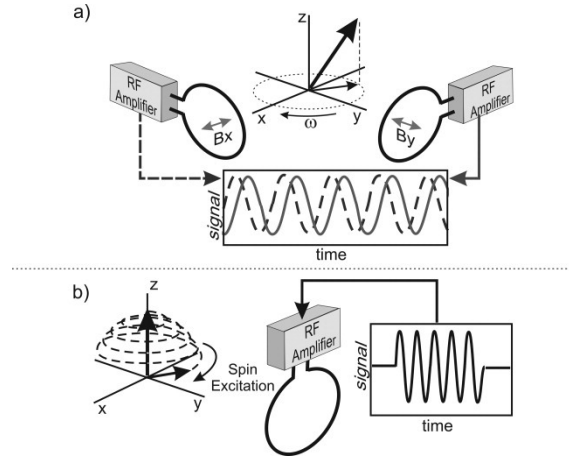


Figure 2.2: A voltage is induced via electromagnetic induction, making an MR signal. (A) The coils are perpendicular and can measure the phase and amplitude of the transverse magnetic vector. (B) RF pulses progressively "flip" the z-axis magnetisation vector into the transverse plane. Figure from [6]

field ( $B_1$ ) must be applied at the Larmor Frequency (the resonance frequency) to flip  $M$  into the transverse plane, which then rotates around  $B_0$  at the Larmor Frequency. This induces a signal in the receiver coils. The  $B_1$  pulses are commonly called radio frequency (RF) pulses since the Larmor frequency is in the radio frequency range. [7][6]

The resulting angle between  $M$  and  $B_0$  is called the flip angle ( $\alpha$ ) and can be calculated using:

$$\alpha = \gamma \int B_1(t) dt \quad (2.2)$$

### 2.1.2 $T_1$ , $T_2$ & $T_2^*$ Relaxation

After an RF pulse is switched off,  $M$  will return to its equilibrium state (precessing with axis aligned with  $B_0$ ). This will cause a reduction in the magnitude of  $M_{xy}$  and, consequently, the MR signal measured; this decrease is known as the Free Induction Decay. The time it takes for the signal to return to its equilibrium state is known as the relaxation time. Two independent relaxation processes affect this time: longitudinal and transverse relaxation.

The longitudinal relaxation process describes the reduction of the flip angle as  $M$  re-aligns with  $B_0$ . The time it takes for  $M_z$  to regain 63% of its original amplitude from a  $90^\circ$  flip angle is called the  $T_1$  relaxation time.  $T_1$  relaxation time is an intrinsic property that can be used to identify a tissue type.[7]

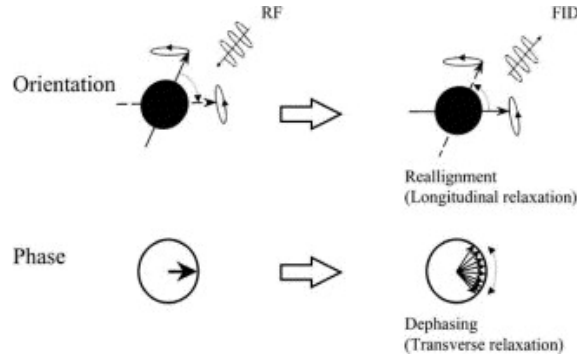


Figure 2.3: **Upper Row:** Longitudinal relaxation (decreasing the flip angle).  
**Lower Row:** Transverse relaxation (spin dephasing) Figure from [7]

The transverse relaxation process describes the loss of signal from proton dephasing. During RF excitation, all protons are in phase. However, protons will have slightly different Larmor frequencies due to inhomogeneities in the  $B_0$  field and local magnetic fields (caused by differing tissue magnetic susceptibilities). The relaxation time for this is called  $T_2'$ . Dephasing is also caused by interactions between neighbouring protons called spin-spin interactions. The relaxation time for this is called  $T_2$ . The dephasing reduces the amplitude of  $M_{xy}$ . The time it takes for the amplitude to reach 37% ( $1/e$ ) of its maximum amplitude is called the  $T_2^*$  relaxation time.

The effects of  $T_2$  can be isolated using a "spin-echo" pulse sequence. Again,  $T_2$  is an intrinsic property that can be used to identify a tissue type and is always shorter than  $T_1$ . [6]

### 2.1.3 Spatial Encoding & k-Space Acquisition

We require extra information to deduce the coordinates of the source of the MR signals captured. To do this, we use magnetic gradient (z-plane), frequency encoding (x-plane), and phase encoding (y-plane).

Since the Larmor frequency is dependent on the magnetic field experienced, a magnetic gradient applied parallel to the  $B_0$  field will create a similar precession frequency gradient along the z-plane. This is then exploited by using a small range of frequencies in the RF pulse to excite (i.e. flip) only a thin slice of protons. Each slice can be selected by using the corresponding band of RF.

After the initial excitation, in which all protons are in phase, a temporary magnetic gradient is applied in the y-plane. In areas with higher magnetisation, this will increase precession frequency and, consequently, angular velocity. Different angular velocities will spread out the phase of protons along the y-axis. When the gradient is removed, they will all continue with the same frequency,

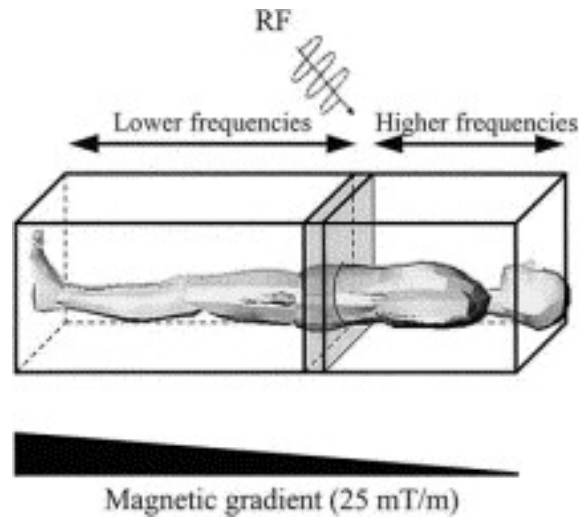


Figure 2.4: Thin slices (3-8mm) can be excited by using a narrow range of RF pulses. Figure from [7]

keeping the range of phases.

For frequency encoding, a magnetic gradient is applied in the x-plane while reading-out the MR signal. This creates a range of frequencies.

Plotting the amplitude of a particular phase and frequency is called K-space. Converting from k-space to image-space is achieved by a reconstruction algorithm that uses Fourier Transforms. However, each image line necessitates an extra signal acquisition cycle, hence creating a trade-off between image accuracy and acquisition time.

#### 2.1.4 Pulse Sequences

A pulse sequence is a description of a series of RF pulses and MR-signal acquisitions. These are "handcrafted by MR experts, who combined strong intuitions and an understanding of spin physics with mathematical solutions of the Bloch equations"[5]. They try to maximise signal-to-noise ratio (SNR) and image contrast while reducing scan times. To do this, they optimise parameters such as the echo time (TE) and repetition time (TR).

For example, the time between two RF pulses is called the repetition time.

### 2.1.5 MRI Simulators & Pulseq

Pulseq is a standardised, open-source framework for defining MRI pulse sequences. Pulseq is hardware-agnostic (compatible with all MR-Linacs).[8]. Pulseq can also be used with many MRI simulators, including both KomaMRI and MRZero.

KomaMRI is a high-performance Julia-based MRI simulator which enables scalable, multi-threaded computation to speed up simulations. [9] MRZero is a "PyTorch-based framework for easy MRI sequence optimisation and development of self-learning sequence development strategies". [10]

### 2.1.6 Magnetic Resonance Fingerprinting (MRF)

Conventional MRI predominantly produces qualitative, "weighted" images where pixel intensity represents a relative contrast. This depends on acquisition parameters such as coil sensitivity, and makes it difficult to standardise across different scanners. In contrast, Quantitative MRI (qMRI) aims to measure the fundamental physical properties of tissues, such as the longitudinal ( $T_1$ ) and transverse ( $T_2$ ) relaxation times mentioned earlier. A significant advancement in this field is Magnetic Resonance Fingerprinting (MRF), which replaces traditional steady-state sequences with "pseudorandomized acquisition parameters". By varying the flip angles and repetition times ( $T_R$ ) throughout the scan, MRF generates a unique signal evolution—or "fingerprint"—for every tissue type. These signals are then pattern-matched against a pre-computed dictionary of Bloch equation simulations, allowing for the simultaneous and rapid mapping of multiple tissue properties in a single acquisition. For MR-guided radiotherapy, this quantitative approach offers the potential for "biological gating", where treatment adaptation is driven by functional changes in tumour physiology rather than just anatomical shifts. [11] [12]

## 2.2 Magnetic Resonance Linear Accelerator (MR-LINAC)

The MR-LINAC represents a paradigm shift in radiation oncology, moving beyond conventional IGRT, which provides only a pre-treatment 'snapshot' of the patient's anatomy. Instead, it enables a continuous feedback loop between real-time imaging and dose delivery, allowing for dynamic adjustment during the treatment itself, known as Adaptive Radiotherapy (ART).

### 2.2.1 Clinical Benefits of ART

By capturing high-contrast images while the patient is on the treatment table, clinicians can account for "inter-fraction" changes (variations between daily sessions) and "intra-fraction" motion (movement during the session, such as breathing or bowel movements).

This real-time visualisation allows for a significant reduction in the Planning Target Volume (PTV). Traditionally, large margins are added to the Clinical Target Volume (CTV) to compensate for uncertainties in organ position. Using MRgRT, these margins can be tightened. In prostate cancer, PTV margins have been reduced to 2mm, significantly reducing the volume of radiated healthy tissue. Further, better soft tissue contrast is essential to accurately delineate the tumour and contour Organs at Risk (OaR) in complex regions like the liver and pancreas.

This allows for "hypofractionation" (delivering a more potent dose over fewer sessions) without increasing the toxicity profile for the patient. This significantly decreases treatment side effects. [13]

### 2.2.2 Current Limitations and Implementation Challenges

Despite its potential, the global adoption of MR-LINAC technology remains in its early stages. As of 2024, four primary operational configurations exist, typically utilising magnetic field strengths of either 0.35T (e.g., ViewRay MRIdian) or 1.5T (e.g., Elekta Unity) [13]. The transition to MRgRT introduces several operational bottlenecks:

- **Complexity and Workflow:** The requirement for "on-table" plan adaptation demands a high level of expertise and significant upskilling for radiation therapists, physicists, and oncologists.
- **Delineation Uncertainty:** While tumours frequently shrink during treatment, it is currently unclear if adapting to these smaller volumes is always safe, as the original tumour bed may still contain microscopic disease [4].
- **Time Constraints:** Re-tuning treatment plans while the patient is immobilised requires rapid image acquisition and contouring to prevent motion artifacts and patient discomfort.

### 2.2.3 The Role of AI and Quantitative MRI

To address the clinical bottleneck of manual contouring, AI-based automated segmentation has emerged as a vital tool. Tools such as the Philips RTdrive Core have demonstrated clinical robustness in prostate cancer patients, significantly reducing the workload for clinicians by automatically outlining the target and OaRs. [14].

Furthermore, the future of MRgRT lies in moving beyond simple anatomical contrast. The integration of quantitative MRI (qMRI), such as measuring tissue cell density through Diffusion-Weighted Imaging, allows for biological targeting. By applying Deep Learning and Reinforcement Learning (RL) to these multi-parametric datasets, the MR-LINAC could eventually provide an end-to-end automated system that optimises dose delivery based on the functional response of the tumour in real-time [4].

## 2.3 Deep Learning in MRI

Long acquisition times represent a major barrier in clinical MRI, primarily due to the inherent trade-off between image quality and speed of imaging. Indeed, the 3D images crucial for tumour delineation require longer acquisition times, which increases the risk of motion artifacts. There is a large body of research into reducing acquisition time while maintaining image quality. Notable breakthroughs include Parallel Imaging (multiple receiver coil arrays) and Compressed Sensing. Unfortunately, Parallel Imaging quickly reaches practical Signal-to-Noise Ratio (SNR) limits, while compressed sensing is prohibitively computationally expensive. [5]

To supplement, DL has been widely applied, both in research and through commercial products, to reduce imaging time and improve image quality. For example, DL is used in post-processing to remove noise, improve image resolution and reduce image inaccuracies (called artifacts) from patient movement or field inhomogeneities (e.g. caused by metal patient implants or pacemakers). [15]

### 2.3.1 Image Acquisition and Reconstruction using Deep Learning

Post-processing operates only on the output image, which allows leveraging traditional imaging techniques (CNN, image kernels) as well as being vendor-neutral (so hardware-agnostic). However, not having the raw signal limits the information available.

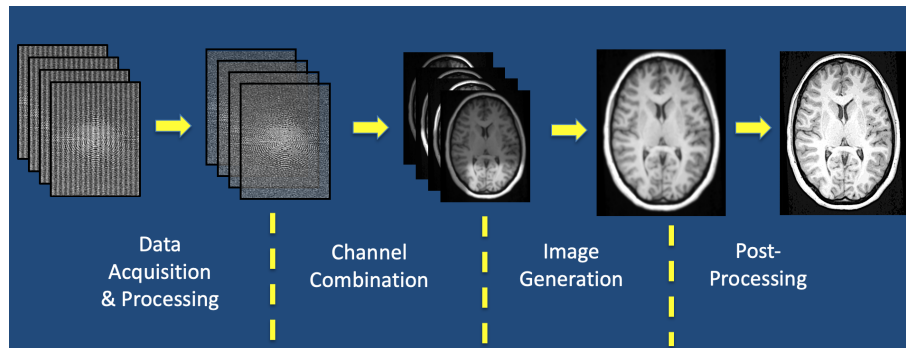


Figure 2.5: Courtesy of Allen D. Elster, MRIquestions.com. [15]

This has given rise to more complex networks that replace both the Fast Fourier Transform algorithm used for reconstruction and post-processing to create higher quality images.

### 2.3.2 Pulse sequence design

Recent research has explored the use of artificial intelligence to optimise the underlying physics of MR acquisition through the autonomous design of pulse sequences. Two differing approaches have been tested: an RL game-based approach and a more traditional supervised learning approach.

The MRzero framework introduces a supervised learning approach that treats the entire scanning and reconstruction pipeline as a fully differentiable system. By simulating RF events, gradient moments, and delay times through a Bloch equation layer, the model can be optimised directly against a target contrast of interest (e.g. a  $T_1$  contrast map). The cost function that was optimised by the model took into account: data fidelity, Specific Absorption Rate (SAR) penalties and scan time. The researchers successfully demonstrated that MRzero could “learn” to replicate conventional gradient-echo sequences from scratch. This was then successfully tested on a clinical 3T scanner for in vivo human brain imaging. This demonstrates that deep learning-derived sequences are clinically viable and can be optimised for specific hardware constraints.[10]

By recasting pulse sequence development as a “game of perfect information,” researchers have implemented a reinforcement learning (RL) framework where an AI agent interacts with a Bloch equation-based physics simulator to generate novel sequence actions. This approach utilises a Bayesian derivative of RL, employing Gaussian processes and Bayesian neural networks to navigate the non-linear dynamics of nuclear magnetisation and maximise an acquisition function based on image reconstruction accuracy. Preliminary 1-D experiments demonstrate that such agents can independently “discover” canonical sequences, such as the gradient echo, and generate novel pulse

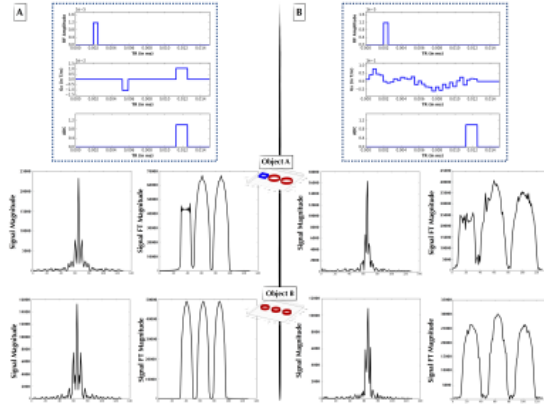


Figure 2.6: (B) Novel pulse sequence successfully developed by AUTOSEQ to mimic a gradient echo sequence. Figure from [16]

sequences that approximate Fourier spatial encoding. While this was a 1-D experiment, further exploration could yield novel pulse sequences that reduce acquisition time and maximise SNR, alleviating the current acquisition bottleneck faced by the MR-Linac.[16]

### 2.3.3 Adaptive MRI acquisition

The above work focuses on the design of novel, but ultimately static pulse sequences. Instead, Adaptive MR is a promising new research direction. Here, a probability distribution of the tissue parameter of interest (e.g.  $T_2$ ) is continuously updated with each new reading. The next pulse sequence is chosen based on the estimate of  $T_2$ . This has been shown to accelerate acquisition time by a factor of 2.5. This is a general paradigm shift in pulse sequence design that motivates and validates further exploration of adaptive MR.[17]

Adaptive MR also has the potential to be combined with RL to autonomously control in real-time MRI hardware. Walker-Samuel demonstrated this by using a Deep Deterministic Policy Gradient (DDPG) algorithm to control an MRI simulation environment where the acquisition process had been turned into a reward-based game. The RL agent was tasked with identifying phantom shapes while being penalised for increased acquisition time. This forces the system to optimise the trade-off between Signal-to-Noise Ratio (SNR) and image acquisition time. Notably, the agent autonomously learned to perform "active sensing," acquiring only four to six outer lines of  $k$ -space and dynamically adjusting  $T_E$ ,  $T_R$ , and flip angles to act as an edge detector. With a reported shape-classification accuracy of 99.8%, this research suggests that RL-controlled scanners could move beyond human-readable imaging to perform high-speed, task-specific "autonomous



sensing”. However, this was purely simulation-based and a very simple task, whether this method can be applied to more complex situations is unproven.

#### 2.3.4 Clinical Implications

This research landscape is the early phases of a transition from human-engineered MRI protocols to a paradigm of ”autonomous radiology.” While the MR-LINAC offers unparalleled soft-tissue visualisation, its clinical utility is currently throttled by the latency of traditional imaging and manual contouring. The integration of Bayesian reinforcement learning [13], end-to-end supervised learning [10], and task-specific autonomous control [18] could provide a technical pathway to overcome these limitations. By treating the MR-LINAC not as a static imaging tool, but as a dynamic, AI-controlled sensor, it becomes possible to collapse the ”imaging-planning-delivery” cycle into a single, synchronised event. This evolution toward an automated, closed-loop system will likely be the catalyst that allows MR-guided radiotherapy to achieve its full potential, providing real-time, biologically adaptive treatments that respond to tumour dynamics.



# 3

## Project Plan & Evaluation

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### 3.1 Project Plan

The project is structured into three distinct phases, transitioning from rapid algorithmic prototyping to high-fidelity validation on clinical hardware. The key challenge will be bridging the gap between Julia (KomaMRI) and Python (PyTorch & RL libraries) and validating the results on the MR-Linac.

Time constraints will also be a big challenge. I will be working on this full-time from the 23rd March to the 12th June, with a 2-week Easter break. This corresponds to 10 full-time weeks, the below plan is based on 12 weeks as I am hoping to make progress on the Environment Setup before

the 23rd March alongside my studies. So far, progress has been limited. I have tested KomaMRI and conducted background research into related works.

### 3.1.1 Phase 1: Prototyping and Environment Setup (Weeks 1–4)

- **Simulator Interfacing:** Develop a Python-based Gymnasium wrapper for the MRzero environment to facilitate reinforcement learning.
- **Digital Phantom Design:** Script a digital twin of the QalibreMD Model 130 phantom, incorporating its 42 spheres with varying  $T_1$ ,  $T_2$ , and PD properties.
- **Milestone:** Successful "sim-to-sim" loop where the RL agent receives a signal from MRzero and proposes a sequence change.

### 3.1.2 Phase 2: RL Training and Simulator Migration (Weeks 5–8)

- **Agent Training:** Train the agent to optimise scan trajectories for the rapid localisation of the Model 130 vials.
- **Simulator Migration:** Port the trained agent to KomaMRI.jl using JuliaCall. This stage tests the agent's robustness when moving from EPG-based physics to Bloch-based physics.
- **Milestone:** Agent converges on an optimised protocol that accurately identifies vial locations in under 5 minutes.

### 3.1.3 Phase 3: Validation and qMRI Extraction (Weeks 9–12)

- **MR-Linac Deployment:** Execute the optimised sequences on the MR-Linac using physical phantoms.
- **Project Write-up:** Complete the final BENG report.
- **Milestone:** Comparison of simulated results against experimental phantom data.

### 3.1.4 Potential Extensions: Advanced qMRI Mapping

If the core objectives are met ahead of schedule, the following extensions will be explored:

- **Autonomous qMRI Mapping:** Beyond mere localisation, the RL reward function will be modified to prioritise the precision of  $T_1$  and  $T_2$  estimates. The agent would "learn" to dwell longer on specific k-space coordinates to maximise the SNR of relaxation time measurements for the QalibreMD spheres.
- **B1+ Inhomogeneity Correction:** Extending the agent's capabilities to identify and compensate for flip-angle variations, a common challenge in 1.5T MR-Linac systems, to improve the accuracy of the resulting qMRI maps.

## 3.2 Evaluation Plan

The project's success will be evaluated through a rigorous comparison of the RL agent's performance against the ground-truth properties of the **QalibreMD System Standard Model 130** phantom. This phantom is specifically designed for the NIST/ISMRM standard and provides traceable reference values for multi-parameter validation.

### 3.2.1 Benchmark: The QalibreMD Model 130 Standard

The evaluation will utilize three primary components of the Model 130 phantom to measure the agent's ability to accurately characterize tissue-mimicking materials:

- **$T_1$  Array:** 14 spheres containing  $\text{NiCl}_2$  with longitudinal relaxation times ranging from  $\sim 20$  ms to 1900 ms.
- **$T_2$  Array:** 14 spheres containing  $\text{MnCl}_2$  with transverse relaxation times ranging from  $\sim 5$  ms to 550 ms.
- **Proton Density (PD) Array:** 14 spheres with PD values ranging from 5% to 100%.

### 3.2.2 Quantitative Performance Metrics

The primary objective is to minimize the "Sim-to-Real" gap. Success will be defined by the following metrics:

| Metric                       | Measurement Description                                                                                           | Success Criteria                              |
|------------------------------|-------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| <b>Parameter Accuracy</b>    | Mean Absolute Percentage Error (MAPE) of $T_1$ , $T_2$ , and PD relative to QalibreMD reference values.           | $< 5\%$ error for all arrays.                 |
| <b>Spatial Localization</b>  | Accuracy in identifying the $(x, y)$ centroids of the 42 spheres within the 5-minute scan window.                 | Error $< 1.0$ mm.                             |
| <b>Scan Efficiency</b>       | Total measurement time required to reach a stable parameter estimate compared to a standard grid-search protocol. | $> 25\%$ reduction in acquisition time.       |
| <b>Signal-to-Noise (SNR)</b> | The agent’s ability to maintain sufficient SNR while adapting sequence parameters in real-time.                   | $\text{SNR} \geq$ standard clinical protocol. |

Table 3.1: Evaluation benchmarks for the RL agent using the Model 130 phantom. Quantitative success criteria are illustrative estimates.

### 3.2.3 Experimental Verification Strategy

1. **Phase I (Sim-to-Sim):** The agent’s ability to “transfer” its learning from the MRzero EPG simulator to the KomaMRI.jl Bloch simulator will be tested. This ensures the RL strategy is robust to more complex physical artifacts (e.g., off-resonance or B1 inhomogeneities).
2. **Phase II (Sim-to-Real):** The final sequence, optimised by the agent, will be deployed on the MR-Linac. The resulting quantitative maps will be processed using standard fitting algorithms (e.g., PhantomViewer or custom Julia scripts) to compare the RL-driven results against the NIST-traceable ground truth.

### 3.2.4 Contribution to the State-of-the-Art

Current qMRI protocols on the MR-Linac often suffer from long acquisition times or motion artifacts. By demonstrating that an RL agent can autonomously navigate the QalibreMD phantom’s parameter space (locating vials and measuring  $T_1/T_2$  simultaneously), this project aims to prove the feasibility of **fully adaptive, autonomous quantitative imaging**.

## 3.3 Legal and Ethical Matters

The development of a reinforcement learning (RL) framework for MRI sequence optimisation poses several legal and ethical considerations.

### 3.3.1 Legal Considerations

This project will build upon MRI simulators and RL libraries, so careful consideration of intellectual property (IP) and software licensing is required. In addition to adhering to the terms of use of the MR-Linac and the phantoms used.

- **Intellectual Property and Open Source:** This project will build exclusively on Open Source software. Using both `KomaMRI.jl` and `MRzero` for MRI simulators, `PulSeq` for sequence specifications, and Python’s extensive ML libraries (e.g. `PyTorch`). All development will adhere to the original MIT or Apache licenses. The resulting code and trained RL agents will be documented and shared under similar open-source terms to promote reproducibility in the scientific community.
- **Regulatory Compliance:** While the project is currently in the research phase, the software will interact with an MR-Linac. Since this is a clinical therapeutic device, development must comply with the manufacturer’s safety protocols. For example, the RL agent will be unable to request gradient switching rates or radio frequency (RF) power levels that exceed hardware safety limits.
- **Data Protection:** This project will only use physical phantoms (e.g. NIST-traceable phantoms) or their digital twins. As such, no Personal Data or Protected Health Information (PHI) will be processed. Consequently, the project does not fall under the jurisdiction of the General Data Protection Regulation (GDPR).

### 3.3.2 Ethical Considerations

While testing will be limited to non-biological subjects, the use of a clinical MR-Linac raises ethical considerations.

- **Safety and Hardware Integrity:** A primary ethical concern is the “black box” nature of reinforcement learning. An agent trained to optimise for speed might explore pulse sequences that are physically taxing on the MRI hardware. To address this, the simulation environment includes hard constraints on the Specific Absorption Rate (SAR) and  $dB/dt$  limits to ensure that the learned strategies are inherently safe for future clinical translation.

- **Data Integrity and Hallucination:** In the context of AI-driven MRI, there is an ethical risk of the agent generating "hallucinated" features or artifacts to maximise its reward function. This project addresses this by using quantitative MRI (qMRI) phantoms with known ground-truth relaxation times ( $T_1, T_2$ ), allowing for a rigorous, objective validation of the agent's accuracy.
- **Professional Responsibility:** The use of the MR-Linac for phantom experiments requires adherence to institutional safety guidelines. All experiments will be conducted under the supervision of qualified medical physicists to ensure that the research does not interfere with clinical workflows or compromise the calibration of equipment used for patient treatment.



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